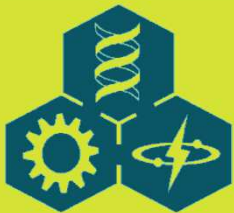
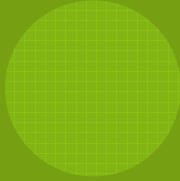


LECTURE 5:

QUANTITATIVE MODEL FORMULATION: II





OUTLINE

5.1 Physical Processes

5.2 Using the Toolbox of Biological Processes

5.3 Useful Functions

5.4 Examples



5.1 PHYSICAL PROCESSES

- ◎ Conservation of Mass and Energy
 - ◎ Biological systems are physical systems that exist in three dimensional space and are subject to fundamental physical laws and process.
 - ◎ The central idea of mass and energy conservation is that material or energy that flows from one place to another is lost from the first and an equal amount is gained by the second place.



5.1 PHYSICAL PROCESSES

🎯 Ecosystem Example

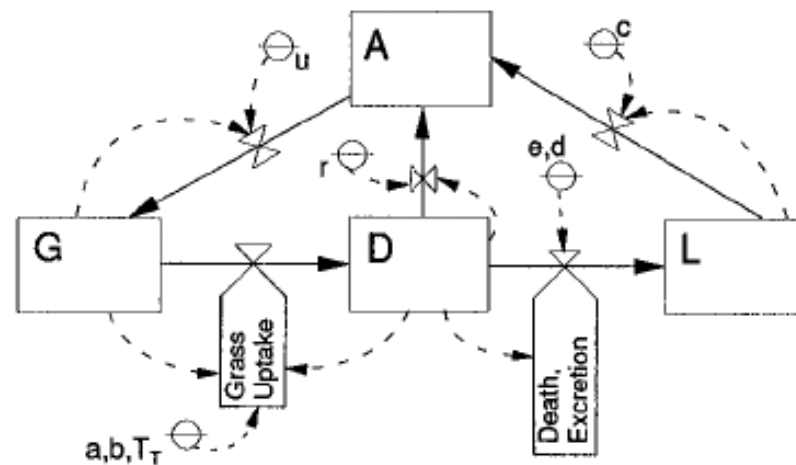
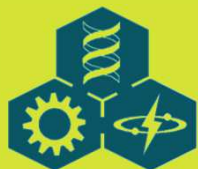


Figure 5.1: Carbon flow in a simple terrestrial ecosystem. A=atmosphere, G=grass, D=deer, L=lumped excretion



5.1 PHYSICAL PROCESSES

◎ Ecosystem Example

$$\frac{dD}{dt} = \left(\frac{aT_r G}{1 + abG} \right) D - rD - D(e + d) \quad (5.1)$$

$$\frac{dG}{dt} = uG - \left(\frac{aT_r G}{1 + abG} \right) D \quad (5.2)$$

$$\frac{dL}{dt} = D(e + d) - cL \quad (5.3)$$

$$\frac{dA}{dt} = -uG + rD + cL \quad (5.4)$$

Table 5.1: Parameter definitions for a carbon flow model.

a	Deer successful search rate for grass
b	Deer handling time while eating grass
c	Rate of decomposition of feces and dead deer by bacteria
d	Rate of feces production by deer
e	Fraction of deer carbon becoming rotting corpses
r	Rate of deer production of gaseous carbon (respiration)
T_r	Total time for foraging
u	Rate of atmospheric carbon uptake by grass



5.1 PHYSICAL PROCESSES

◎ Spatial Flows

- ◎ When the spatial resolution is such that only a few, large regions are modeled (such as broad areas in a lake), then the problem can be treated just as we treated the carbon flow problem. We write *ordinary differential* for each spatial area with appropriate flows between the various spatial regions.
- ◎ In situations where we can not reasonably assume homogeneous regions (i.e., where there is a continuous gradation of the spatial structure), we must use a different conceptual framework. In these cases, the framework we use is based on *partial differential equations (PDEs)*.



5.1 PHYSICAL PROCESSES

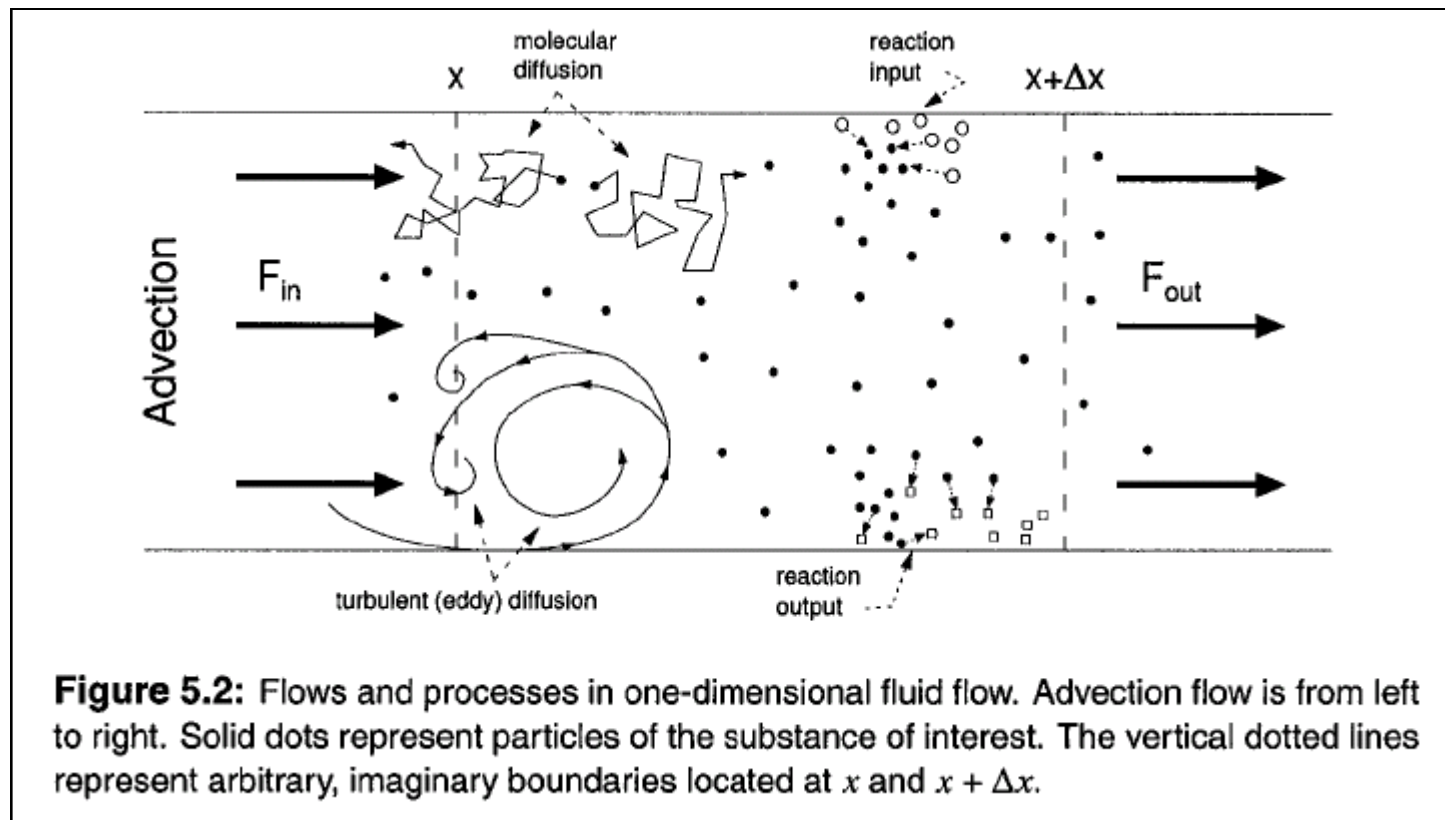
◎ Spatial Flows

- ◎ Four fundamental processes affecting fluids and solutes recur in models which have two independent variables (time and space) over which the state variable (C) varies:
 1. Advection
 2. Molecular diffusion
 3. Turbulent diffusion
 4. Reaction.



5.1 PHYSICAL PROCESSES

◎ Spatial Flows



5.1 PHYSICAL PROCESSES

⊙ Spatial Flows

- ⊙ Discrete formulation of ordinary differential equation:

$$y_{t+\Delta t} = y_t + \Delta t [NetChange(t)] \quad (5.5)$$

- ⊙ Finite difference equation based on discretized space:

$$\begin{aligned} C_{out} &= C_{in} - \Delta x [NetChange(x)] \\ C_x &= C_{x+\Delta x} + \Delta x [NetChange(x)] \end{aligned} \quad (5.6)$$

$$\lim_{\Delta x \rightarrow 0} \frac{C_x - C_{x+\Delta x}}{\Delta x} = -\frac{dC}{dx}$$

- ⊙ When we add time and require conservation of mass, we must insure that the temporal changes in C equal the spatial changes in C.

$$\frac{\partial C}{\partial t} = -\frac{\partial F}{\partial x} \quad (5.7)$$



5.1 PHYSICAL PROCESSES

◎ Spatial Flows

- ◎ The rate of change of the concentration in a segment must equal the inflow minus the outflow.

$$F_{out} = F_{x+\Delta x} = F_x + \frac{\partial F}{\partial x} \Delta x$$

- ◎ The statement of the principle of conservation of mass:

$$\Delta M = A \left[\underbrace{F_x}_{F_{in}} - \underbrace{\left(F_x + \frac{\partial F}{\partial x} \Delta x \right)}_{F_{out}} \right] \Delta x$$



5.1 PHYSICAL PROCESSES

◎ Spatial Flows

- ◎ **Advection** : Advection is the flow of media and the solute from point to point. If the velocity is a constant U over a small spatial interval, then the flux of C is simply

$$F = UC$$

by conservation of mass

$$\frac{\partial C}{\partial t} = -\frac{\partial F}{\partial x} = -\frac{\partial(UC)}{\partial x} \quad (5.8)$$



5.1 PHYSICAL PROCESSES

⊙ Spatial Flows

- ⊙ **Diffusion:** Molecular diffusion is the movement of mass due to random motion of individual molecules. D is a constant called *diffusivity* and is assumed here to be constant over x .

$$F = D \frac{\partial C}{\partial x} \quad (5.9)$$

$$\frac{\partial C}{\partial t} = -\frac{\partial F}{\partial x} = -\frac{\partial D \frac{\partial C}{\partial x}}{\partial x} = D \frac{\partial^2 C}{\partial x^2}$$

- ⊙ Putting advection and diffusion together to find changes in C , we have the *conservation equation*:

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2} - \frac{\partial(UC)}{\partial x} \quad (5.10)$$



5.1 PHYSICAL PROCESSES

⊙ Spatial Flows

- ⊙ **Reactions:** Reaction processes are any processes other than advection and diffusion that change the concentration of a solute inside the spatial interval Δx . These may be chemical interactions, or biological uptake and excretion.
- ⊙ An example of *reaction-diffusion equation*:

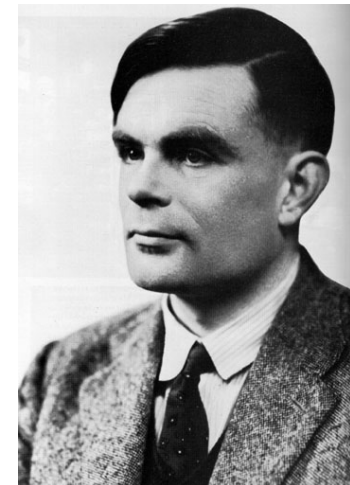
$$\frac{\partial N}{\partial t} = \underbrace{D \frac{\partial^2 N}{\partial x^2}}_{\text{diffusion}} - \underbrace{\frac{\partial(UN)}{\partial x}}_{\text{advection}} - \underbrace{\mu_{\max} \frac{NP}{K_m + N}}_{\text{uptake}} + \underbrace{eS}_{\text{excretion}} \quad (5.11)$$



5.1 PHYSICAL PROCESSES

- ◎ Turing system
 - ◎ Alan Turing, 1952
 - ◎ A diffusion reaction system
 - ◎ Two or more chemical species
 - ◎ Different rate of diffusion for the participants
 - ◎ Chemical interactions

$$\frac{\partial u}{\partial t} = D_u \nabla^2 u + f(u, v)$$
$$\frac{\partial v}{\partial t} = D_v \nabla^2 v + g(u, v)$$

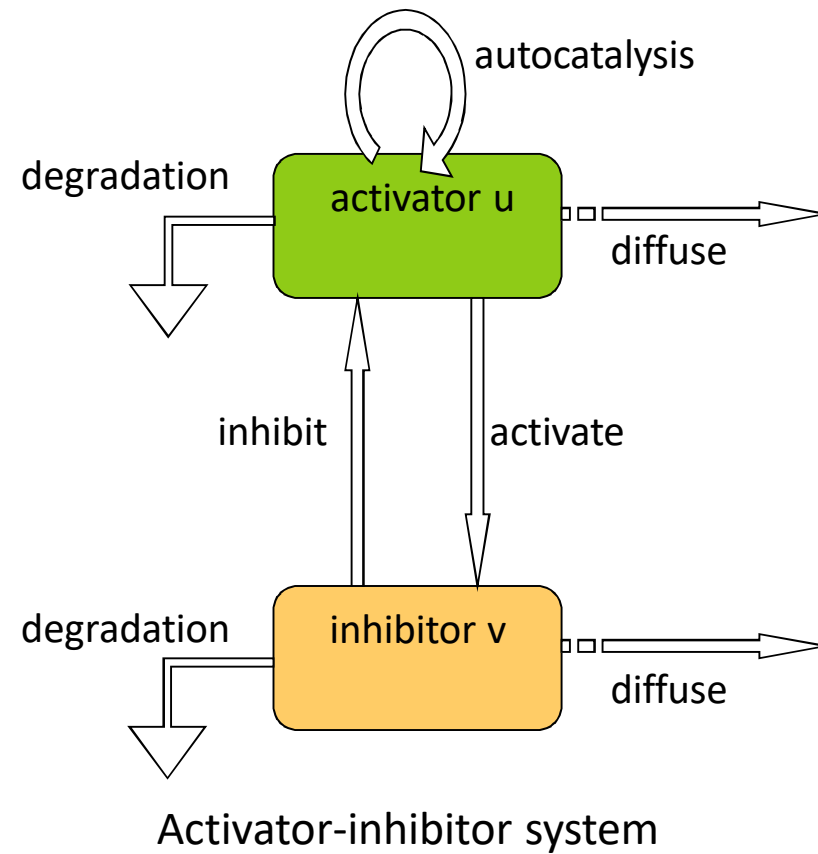


5.1 PHYSICAL PROCESSES

◎ Turing system

$$\frac{\partial u}{\partial t} = D_u \nabla^2 u + f(u, v)$$

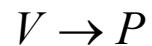
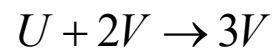
$$\frac{\partial v}{\partial t} = D_v \nabla^2 v + g(u, v)$$



5.1 PHYSICAL PROCESSES

- ⊙ Turing system
 - ⊙ Gray-Scott Model

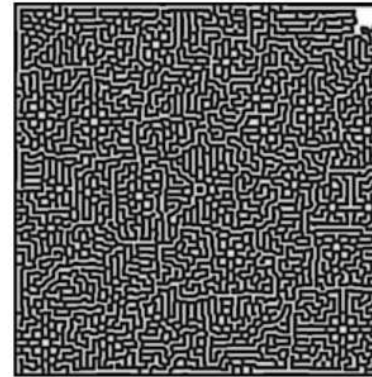
Chemical Reaction



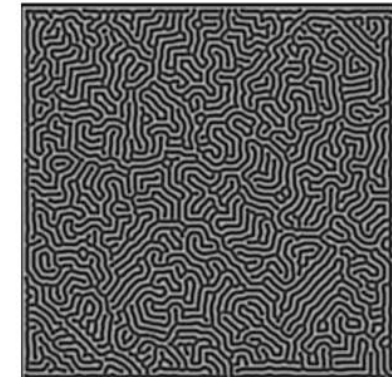
Model Equations

$$\frac{\partial u}{\partial t} = D_u \nabla^2 u - uv^2 + F(1 - u)$$

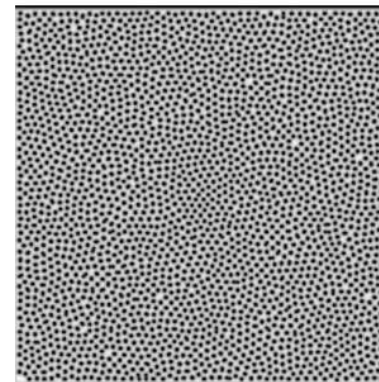
$$\frac{\partial v}{\partial t} = D_v \nabla^2 v + uv^2 - (F + k)v$$



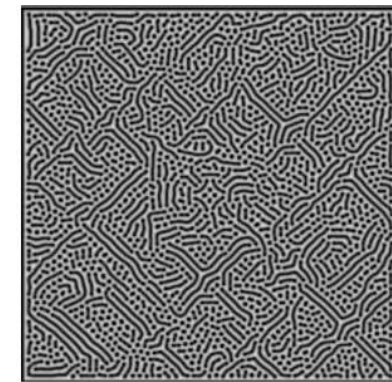
Flower-pattern: $F=0.055$; $k=0.062$



Mazes-pattern: $F=0.029$; $k=0.057$



Mitosis-pattern: $F=0.028$; $k=0.062$



Solitons-pattern: $F=0.03$; $k=0.06$

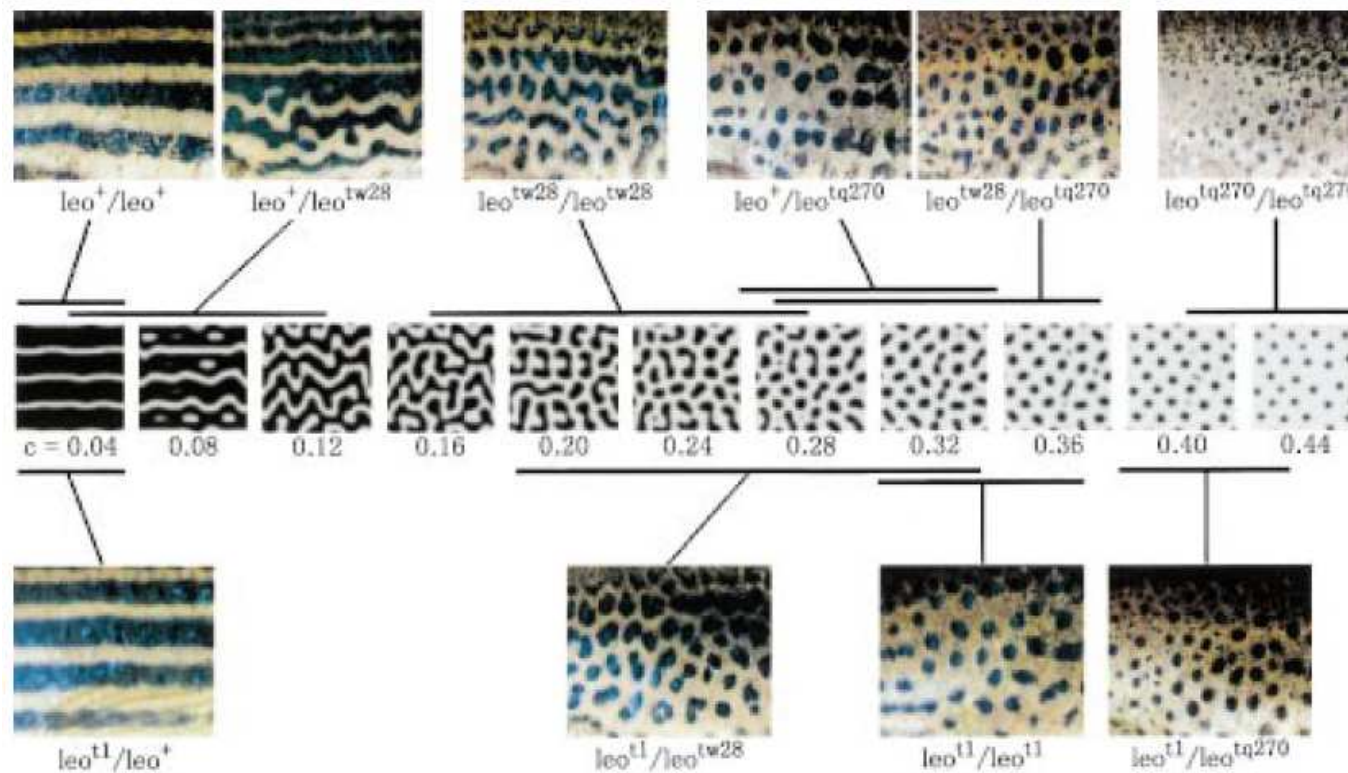
Reference and Simulation Link:

https://itp.uni-frankfurt.de/~gros/StudentProjects/Projects_2020/projekt_schulz_kaefer/



5.1 PHYSICAL PROCESSES

◎ Pattern formation modeling of zebrafish

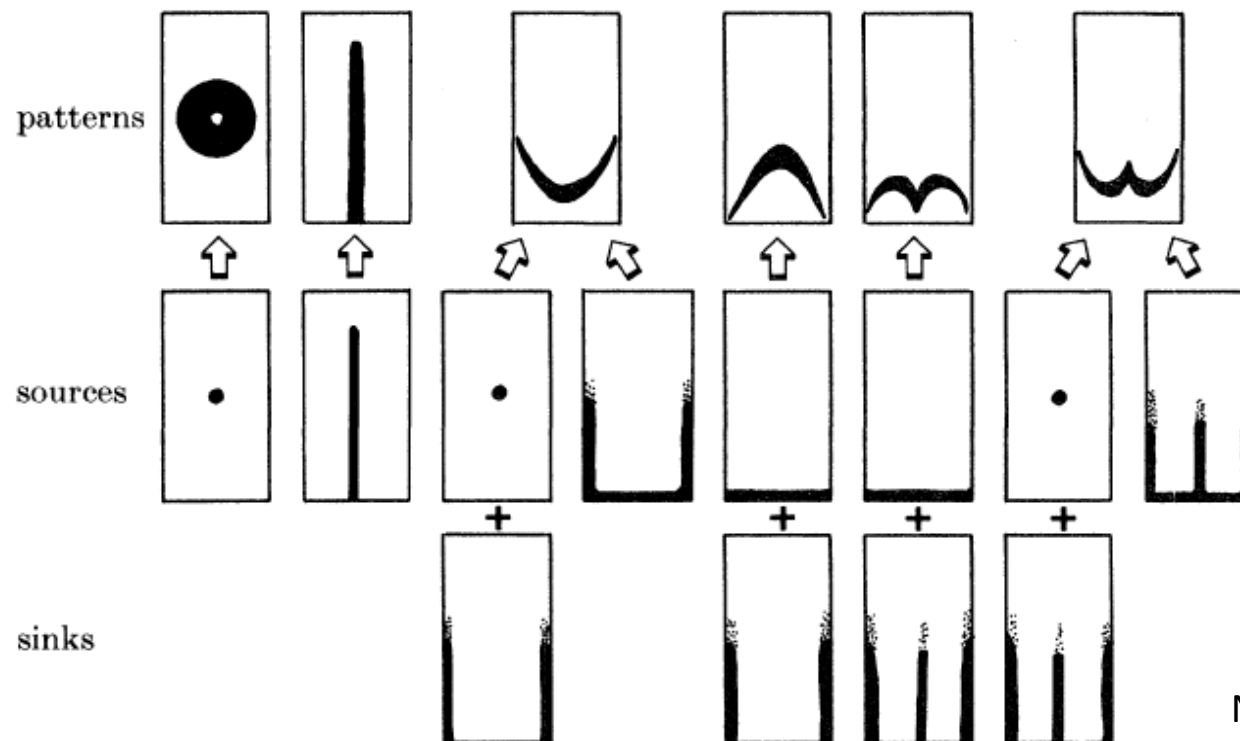


Asai et al., 1999



5.1 PHYSICAL PROCESSES

◎ Simulation of butterfly wing pattern elements

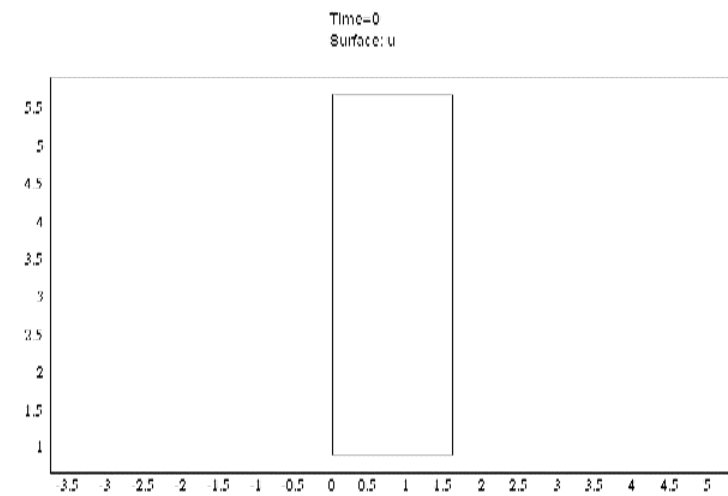
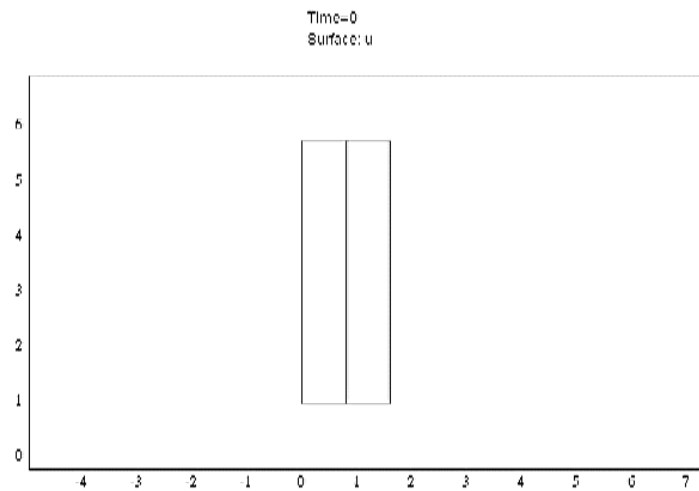


Nijhout, 1991



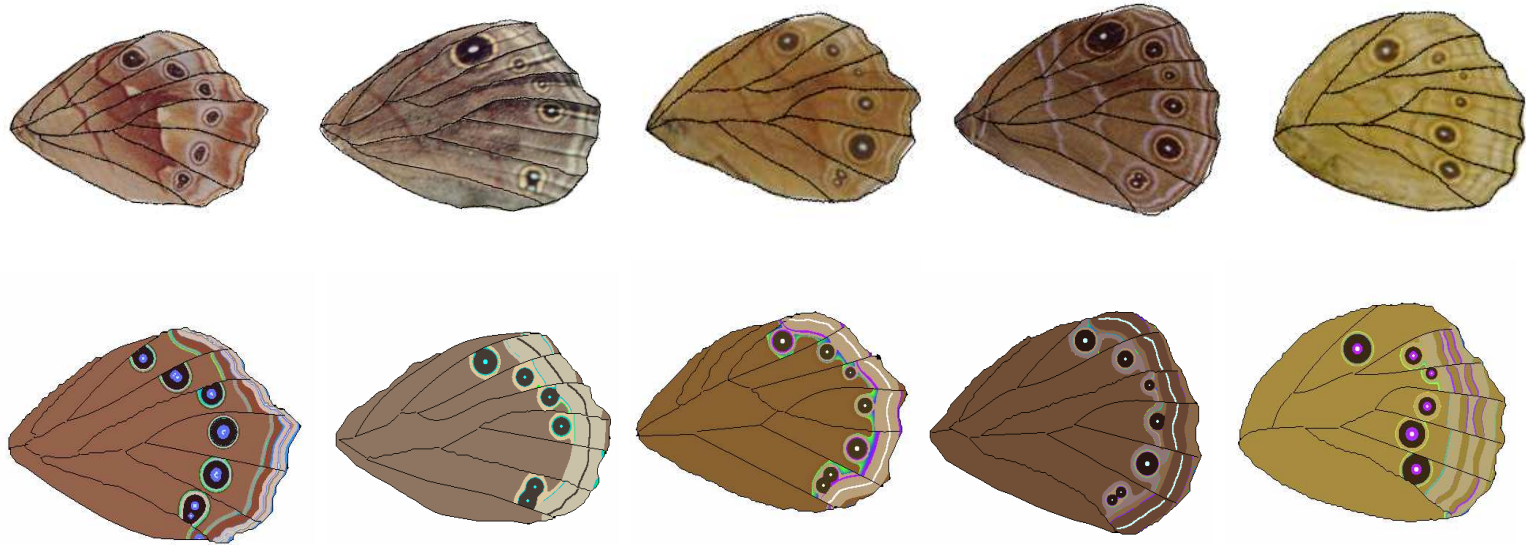
5.1 PHYSICAL PROCESSES

◎ Simulation of butterfly wing pattern elements



5.1 PHYSICAL PROCESSES

◎ Simulation of eyespots in wing veins



5.1 PHYSICAL PROCESSES

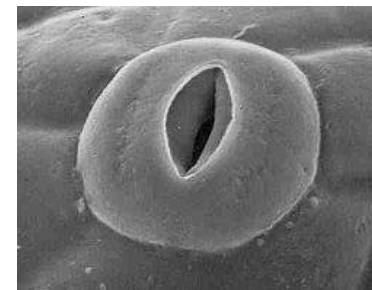
⊙ Discontinuous Functions

- ⊙ For reasons of simplicity and convenience if nothing else, we often choose to represent the phenomena as discontinuous. A hypothetical example is

$$R = \begin{cases} 2x & \text{if } 0 \leq x < 0.5 \\ 1.0 & \text{if } 0.5 \leq x < 1.0 \\ 1.0 - bx & \text{if } 1.0 \leq x \end{cases}$$

- ⊙ An example of describing the opening of the stomata on the leaf surface.

$$a = \begin{cases} b_g P_g - b_e P_e & \text{if } b_g P_g > b_e P_e \\ 0.0 & \text{otherwise} \end{cases}$$



5.1 PHYSICAL PROCESSES

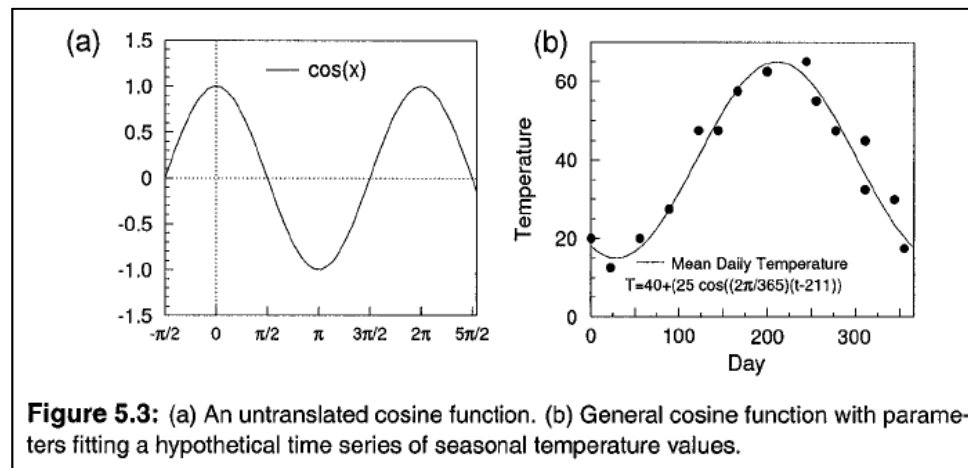
⊙ Time and Driving Variables

- ⊙ Driving variables described by a time-dependent equation:

$$y = M + A \cos(\omega(t - t_0)) \quad (5.12)$$

$$T = 40 + 25 \cos\left(\frac{2\pi}{365}(t - 211)\right)$$

- ⊙ Incorporating time in functions used in computer simulations using *look-up table*.



5.2 USING THE TOOLBOX OF BIOLOGICAL PROCESSES

- ◎ Basic Tools in our Toolbox for Reading and Constructing:
 1. Constant rates
 2. Relative rates
 3. Feedback
 4. Mass action
 5. Conservation of mass
 6. Limitation by multiple controls
 7. Discontinuous functions
 8. Time dependence
- ◎ An approach to successfully reading and constructing quantitative models is to combine these basic formulations in ways that represent the biological hypotheses.



5.2 USING THE TOOLBOX OF BIOLOGICAL PROCESSES

☉ Checking Units

- ☉ The physical units of the derivative must match the units of the equation on the right-hand side.
- ☉ This will check for two types of errors:
 1. Inappropriate expressions (e.g., dividing when you should subtract).
 2. Bad logic that requires parameter values with incorrect units.
- ☉ Example: logistic equation (Eq. 4.14)

$$\frac{dN}{dt} = \left[r \left(1 - \frac{N}{K} \right) \right] N = rN - \frac{r}{K} N^2 \quad (4.14)$$

$$\frac{\text{numbers}}{\text{time}} = \frac{1}{\text{time}} \left(\text{unitless} - \frac{\text{numbers}}{\text{numbers}} \right) \text{numbers} = \frac{\text{numbers}}{\text{time}}$$



5.2 USING THE TOOLBOX OF BIOLOGICAL PROCESSES

◎ Conversion to Dimensionless Format

- ◎ A useful procedure reduces the number of parameters by converting the differential equation to a dimensionless form, thereby creating new variables and parameters, but also eliminating many old variables and parameters.
- ◎ The net gain is fewer parameters.
- ◎ The objective, then, is to manipulate the equation to replace all parameters and variables with dimensionless quantities.



5.2 USING THE TOOLBOX OF BIOLOGICAL PROCESSES

◎ Conversion to Dimensionless Format

◎ Example:

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K} \right) \quad (5.13)$$

$$\frac{d(\bar{N}\check{N})}{d(\check{t}\bar{t})} = r\bar{N}\check{N} \left(1 - \frac{\bar{N}\check{N}}{K} \right) \quad \leftarrow \text{create unitless variables}$$

$$\frac{d(\bar{N}\check{N})}{d(\check{t})} = \check{t} r \bar{N} \check{N} \left(1 - \frac{\bar{N}\check{N}}{K} \right) \quad \leftarrow \text{multiply by } \check{t}$$

$$\frac{d(\bar{N})}{d(\check{t})} = \check{t} r \bar{N} \left(1 - \frac{\bar{N}\check{N}}{K} \right) \quad \leftarrow \text{divided by } \check{N} \quad (5.14)$$

$$\frac{d(\bar{N})}{d(\check{t})} = \bar{N} (1 - \bar{N}) \quad \leftarrow \text{define } \check{t} \text{ and } \check{N} \quad (5.15)$$

where the new quantities are $\check{t} = 1/r$ and $\check{N} = K$.

We have reduced the number of parameters from 2 to 0,

and we have essentially scaled time by $1/r$ and population size by K .



5.2 USING THE TOOLBOX OF BIOLOGICAL PROCESSES

◎ Conversion to Dimensionless Format

◎ Mechanical Steps:

1. Make a table of the state variables and parameters and their units.
2. Re - write the differential equations, substituting for each state variable a product of a dimensionless scaling variable ($\bar{x}$) and a variable representing 1 unit of that variable ($\check{x}$). E.g., if x is measured in gmC/liter, then for every occurrence of x in the original equations, write :

$$x = \bar{x}\check{x}$$

E.g., a single, linear ODE would be :

$$\frac{dx}{dt} = ax$$

yields :

$$\frac{d\bar{x}\check{x}}{d\bar{t}\check{t}} = a\bar{x}\check{x} \quad (5.16)$$



5.2 USING THE TOOLBOX OF BIOLOGICAL PROCESSES

◎ Conversion to Dimensionless Format

◎ Mechanical Steps:

3. Make the left - hand - side of the ODES unitless by multiplying both sides by $\tilde{t} / \tilde{x}$. Do this for all differential equations before proceeding. Cancel any $\tilde{x}$ possible. For example, Eq. 5.16 becomes :

$$\frac{d\tilde{x}}{d\tilde{t}} = a\tilde{t}\tilde{x}$$



5.2 USING THE TOOLBOX OF BIOLOGICAL PROCESSES

◎ Conversion to Dimensionless Format

◎ Creative Steps:

The goal of this step is to substitute the definition of the new parameters into the modified equations to obtain the dimensionless equations with fewer parameters.

4. Collect the terms with units together in the equations.

5. $\tilde{t}$ will generally be easier to define than the $\tilde{x}$, so try to define the latter first.

6. If $\tilde{x}$ appears as the only variable in one of the components of the equation, then use that component to define $\tilde{x}$. For example, if

$$\frac{d\bar{y}}{d\bar{t}} = \tilde{t} K \tilde{x} \bar{x} + \tilde{t} Q \tilde{x}^2 \bar{y} \bar{x} \bar{y}$$

from the first component on the right define

$$\tilde{x} = \frac{1}{\tilde{t} K}$$

Substitute the new definition of $\tilde{x}$ in to the second component so that $\bar{y}$, now stands alone. Use the same logic to define it as you did $\tilde{x}$ earlier.



5.2 USING THE TOOLBOX OF BIOLOGICAL PROCESSES

◎ Conversion to Dimensionless Format

◎ Creative Steps:

7. Complicated algebraic expressions involving $\tilde{x}$ should be simplified as much as possible before trying definitions. For example, simplify the Michaelis - Menten component of the chemostat model to

$$V_{\max} \left[\frac{\bar{N}}{\frac{K_m}{\bar{N}} + \bar{N}} \right]$$

before defining $\tilde{N}$.

8. It is not a good idea to define one variable ($\tilde{y}$) in terms of another ($\tilde{x}$): use only constants and $\tilde{t}$.



5.2 USING THE TOOLBOX OF BIOLOGICAL PROCESSES

◎ An Example without a Biological Interpretation

$$\begin{aligned}\frac{dx}{dt} &= ax^3 + by^2 \\ \frac{dy}{dt} &= cy^2 - fxy\end{aligned}\tag{5.17}$$

Variable	Units
a	$1/(t \cdot x^2)$
b	$x/(t \cdot y^2)$
c	$1/(t \cdot y)$
f	$1/(t \cdot x)$
x	Unspecified
y	Unspecified



5.2 USING THE TOOLBOX OF BIOLOGICAL PROCESSES

◎ An Example without a Biological Interpretation

Making the left side non - dimensional yields :

$$\frac{d\bar{x}}{d\bar{t}} = \bar{t} a \bar{x}^2 \bar{x}^3 - \frac{b\bar{t}}{\bar{x}} \bar{y}^2 \bar{y}^2 \quad (5.18a)$$

$$\frac{d\bar{y}}{d\bar{t}} = \bar{t} c \bar{y} \bar{y}^2 - f \bar{t} \bar{x} \bar{x} \bar{y} \quad (5.18b)$$

define :

$$\bar{x} = \frac{1}{f\bar{t}} \quad \text{and} \quad \bar{y} = \frac{1}{c\bar{t}} \quad (5.19)$$

after substitution :

$$\frac{d\bar{y}}{d\bar{t}} = \bar{y}^2 - \bar{x} \bar{y}$$

$$\frac{d\bar{x}}{d\bar{t}} = \frac{a}{f^2 \bar{t}} \bar{x}^3 - \frac{bf}{c^2} \bar{y}^2$$



5.2 USING THE TOOLBOX OF BIOLOGICAL PROCESSES

◎ An Example without a Biological Interpretation

The first component on the right side, however, still has $\tilde{t}$ which needs to be defined in terms of constants and chosen to eliminate parameters. Defining $\tilde{t}$ as :

$$\tilde{t} = \frac{a}{f^2}$$

The final non - dimensional equations are :

$$\begin{aligned}\frac{d\bar{x}}{d\tilde{t}} &= \bar{x}^3 - a_1 \bar{y}^2 \\ \frac{d\bar{y}}{d\tilde{t}} &= \bar{y}^2 - \bar{x}\bar{y}\end{aligned}\tag{5.20}$$

$$\text{where} \quad \bar{x} = \frac{f}{a} \quad \bar{y} = \frac{f^2}{ac} \quad a_1 = \frac{bf}{c^2}\tag{5.21}$$

which reduces the number of parameters from 4 to 1, and substituting the units from the above table, shows a_1 to be unitless.



5.2 USING THE TOOLBOX OF BIOLOGICAL PROCESSES

- ◎ Scaling Dimensionless Quantities
 - ◎ Once the non-dimensional equations are derived, we need to provide some interpretations of the constants. Often these provide insight into the processes of interest.
 - ◎ Non-dimensionalization has revealed to us the basic form of the equation. We can recover all the other equation of the same form that might apply to a particular model (e.g. population model) by stretching or shrinking our dimensionless time and state variable.



5.2 USING THE TOOLBOX OF BIOLOGICAL PROCESSES

◎ Buckingham Pi

- ◎ In 1914, Edgar Buckingham proved a theorem that says: *Given a physical relationship with P parameters and D dimensional units, the number of independent dimensionless groups is $P - D$.* In other words, if the original model has P parameters, it can be reduced, without changing the mathematical behavior, to a model with $P - D$ parameters.
- ◎ Knowing the number of variables in the problem and the number of fundamental units (clearer in physical problems than biological ones), the Buckingham Pi Theorem states we can write the model using $P - D$ independent parameters.



5.2 USING THE TOOLBOX OF BIOLOGICAL PROCESSES

◎ Conservation Principle

- ◎ If a model uses a conserved quantity (e.g., g C) all of whose sources and sinks are accounted for, then a state variable can be eliminated from the system of equations. Suppose a fixed amount K of carbon flows among three state variables (x_i), each described by an ODE. Since $K = x_1 + x_2 + x_3$, and K is a constant, we can rewrite any one of the x_i in terms of the other state variables and total C: $x_3 = K - x_1 - x_2$. x_3 effectively becomes an auxiliary variable and we can substitute $K - x_1 - x_2$ anywhere x_3 is used.



5.2 USING THE TOOLBOX OF BIOLOGICAL PROCESSES

◎ Rule of Thumb

◎ Some maxims of model formulation

1. Know the purpose.
2. Know the question.
3. Understand the objects.
4. Reconcile the diagram with the rate equation.
5. Check the units.
6. Extrapolate the functions.
7. Simplify the model.



5.2 USING THE TOOLBOX OF BIOLOGICAL PROCESSES

◎ Rule of Thumb

◎ Techniques for model simplification:

1. Reduce the equations to dimensionless variables.
2. Aggregate state variables.
3. Exploit conservation principles.
4. Use linear functions initially.
5. Use descriptive, phenomenological representations before detailed, mechanistic processes. When objectives or model failure require it, increase the level of details.
6. Assume homogeneous space.



5.3 USEFUL FUNCTIONS

◎ Introduction

- ◎ Many of the biological processes can be represented by a variety of equations (e.g., hyperbolic saturation as either Michaelis-Menten or Holling disc equation). Some are nearly identical in shape, but use different parameters.
- ◎ Choosing among these, unless there are theoretical reasons, is largely a matter of taste and the appropriateness of the normal interpretation of the parameters.



5.3 USEFUL FUNCTIONS

Linear

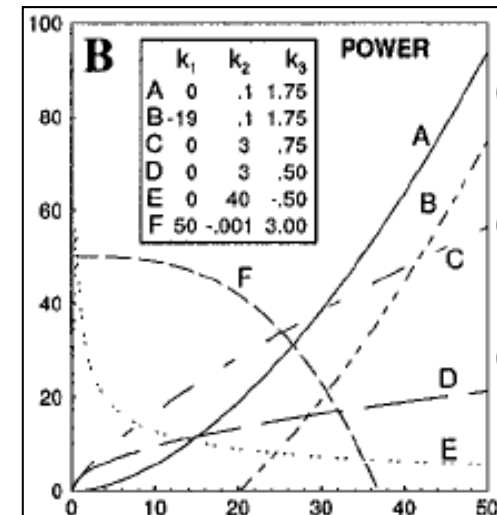
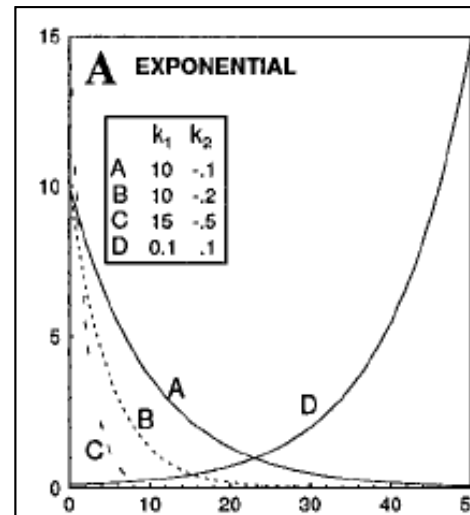
$$y = k_1 + k_2x$$

Exponential

$$y = k_1 e^{k_2 x}$$

Power

$$y = k_1 + k_2 x^{k_3}$$

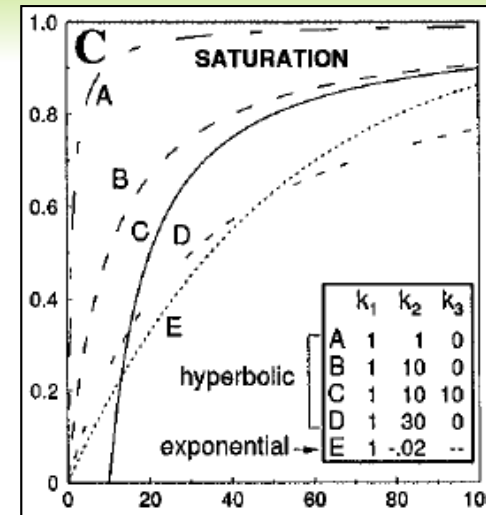


5.3 USEFUL FUNCTIONS

☉ Saturation

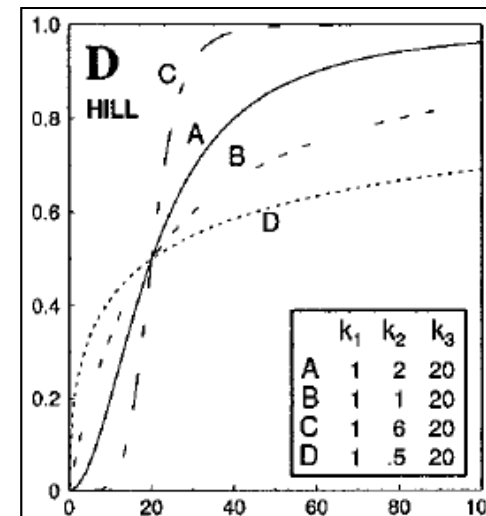
$$y = k_1 \left(\frac{(x - k_3)}{k_2 + (x - k_3)} \right)$$

$$y = k_1 (1 - e^{-k_2 x})$$



☉ Hill

$$y = k_1 \left(\frac{x^{k_2}}{k_3^{k_2} + x^{k_2}} \right)$$



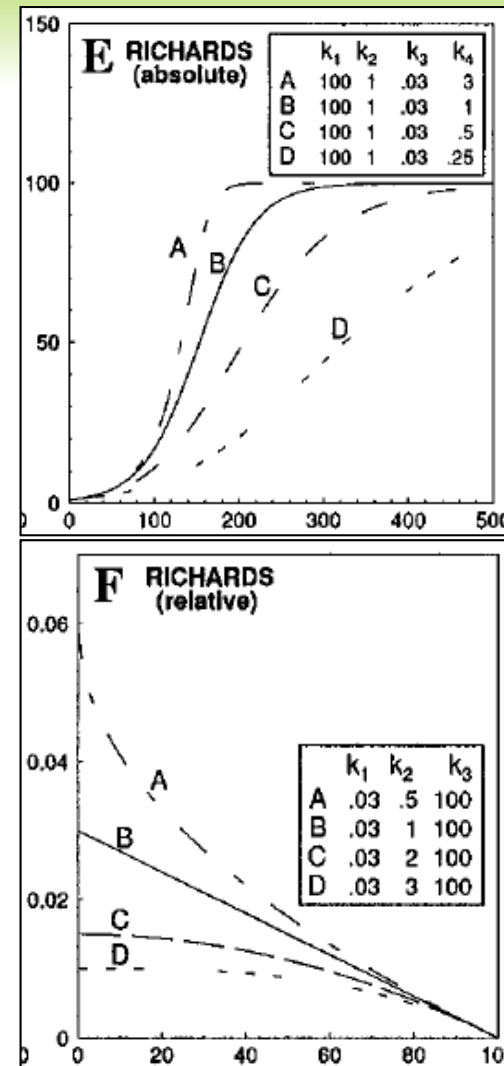
5.3 USEFUL FUNCTIONS

Richards Absolute

$$y = \frac{k_1}{\left(1 + \left(\frac{k_1}{k_2} - 1\right)e^{-k_3 k_4 x}\right)^{1/k_4}}$$

Richards Relative

$$y = \frac{k_1}{k_2} x \left[1 - \left(\frac{x}{k_3}\right)^{k_2} \right]$$



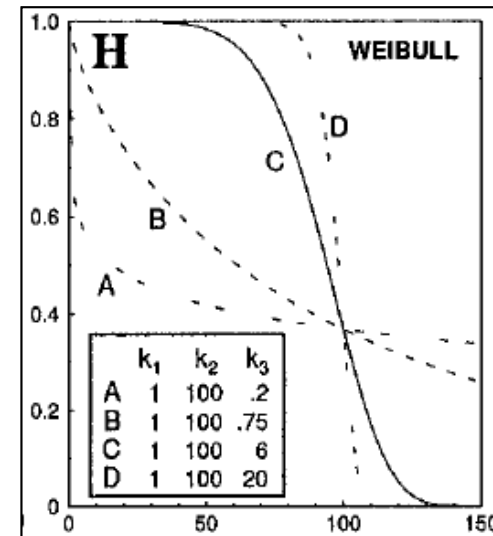
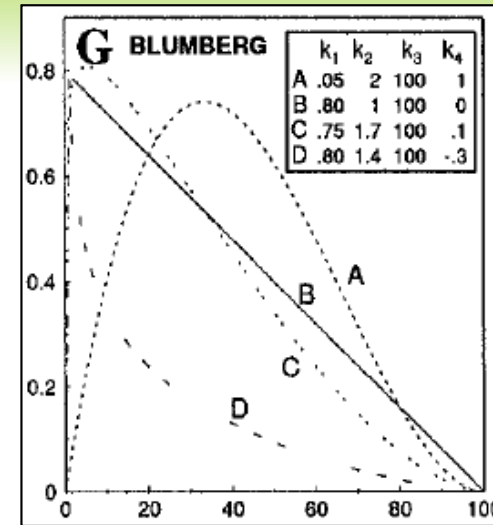
5.3 USEFUL FUNCTIONS

Blumberg

$$y = k_1 x^{k_4} \left[1 - \left(\frac{x}{k_3} \right)^{k_2} \right]$$

Complemented Weibull

$$y = k_1 \exp \left(- \left[\frac{x}{k_2} \right]^{k_3} \right)$$



5.3 USEFUL FUNCTIONS

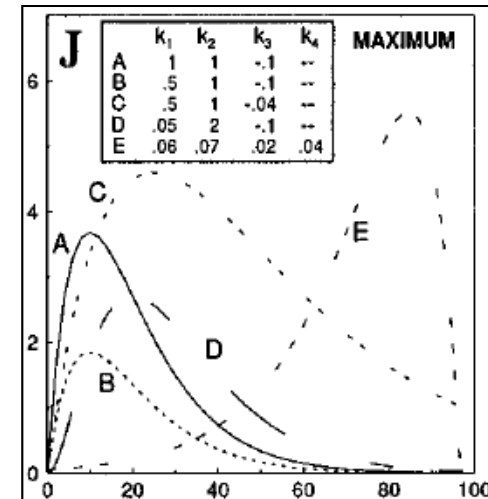
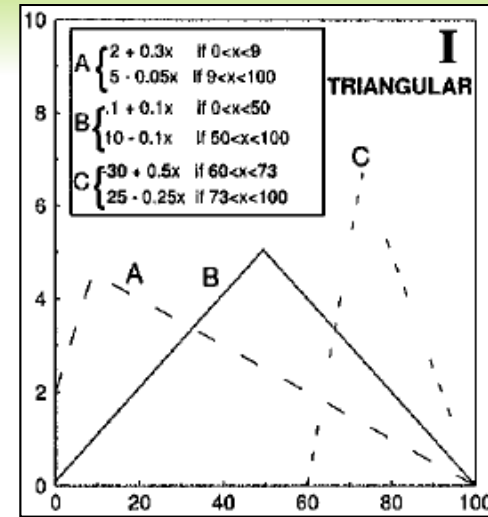
Triangular

$$y = \begin{cases} k_1 + k_2x & \text{if } x < k_3 \\ k_4 - k_5x & \text{if } x \geq k_3 \end{cases}$$

Maxima

$$y = k_1 x^{k_2} e^{k_3 x}$$

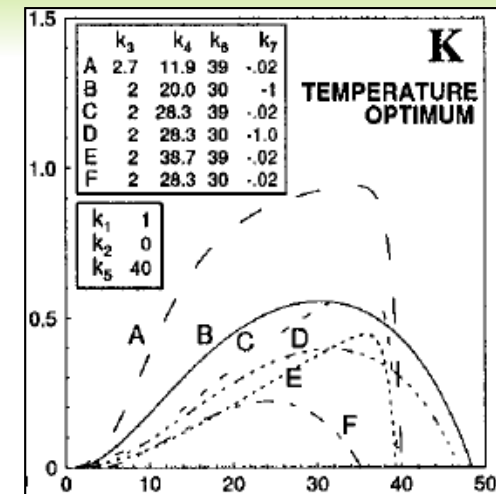
$$y = k_1 e^{k_2 x} (1 - k_3 e^{k_4 x})$$



5.3 USEFUL FUNCTIONS

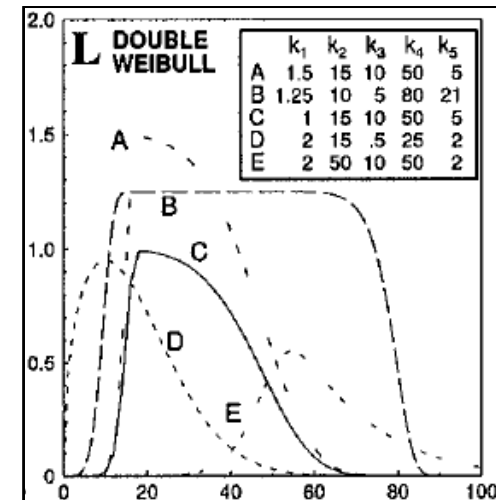
Temperature Optimum

$$y = \frac{k_1(x - k_2)^{k_3}}{k_4^{k_3} + (x - k_2)^{k_3}} - \exp\left(k_7 - \left(\frac{k_5 - (x - k_2)}{k_5 - k_6}\right)\right)$$



Double Weibull

$$y = k_1 \left(1 - e^{-(x/k_2)^{k_3}}\right) e^{-(x/k_4)^{k_5}}$$



5.3 USEFUL FUNCTIONS

◎ Trigonometric

$$y = \sum_{i=1}^N M_i + A_i \cos(\omega_i(x - x_0))$$

◎ Cubic Spline

Cubic splines is such a method that uses a third order polynomial for each subset of the data and smoothly joins the separate cubic equations together.

$$y = k_0 + k_1x + k_2x^2 + \dots + k_nx^n$$

◎ Polynomial

$$y = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

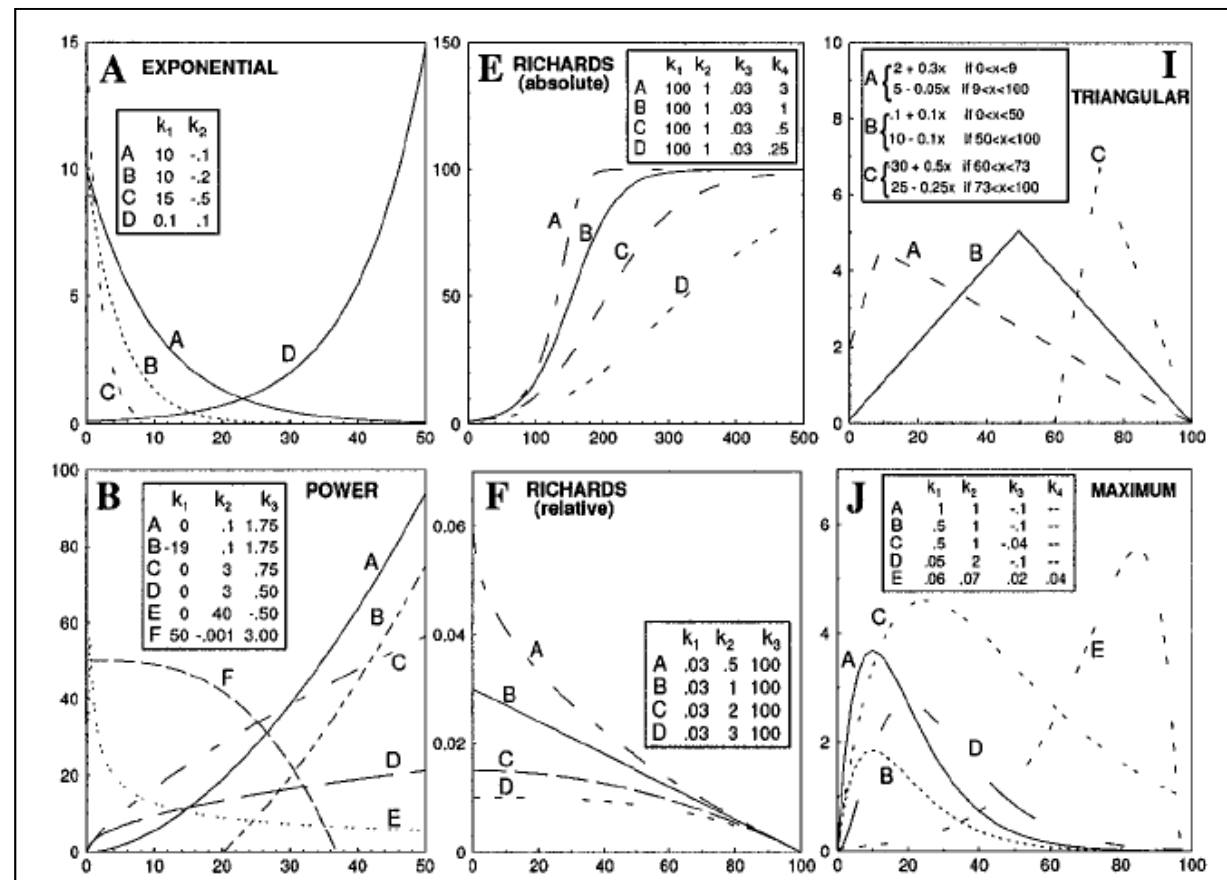
◎ Rational Functions

$$y = \frac{a_0 + a_1x + a_2x^2 + \dots + a_nx^n}{1 + b_0 + b_1x + b_2x^2 + \dots + b_nx^n}$$



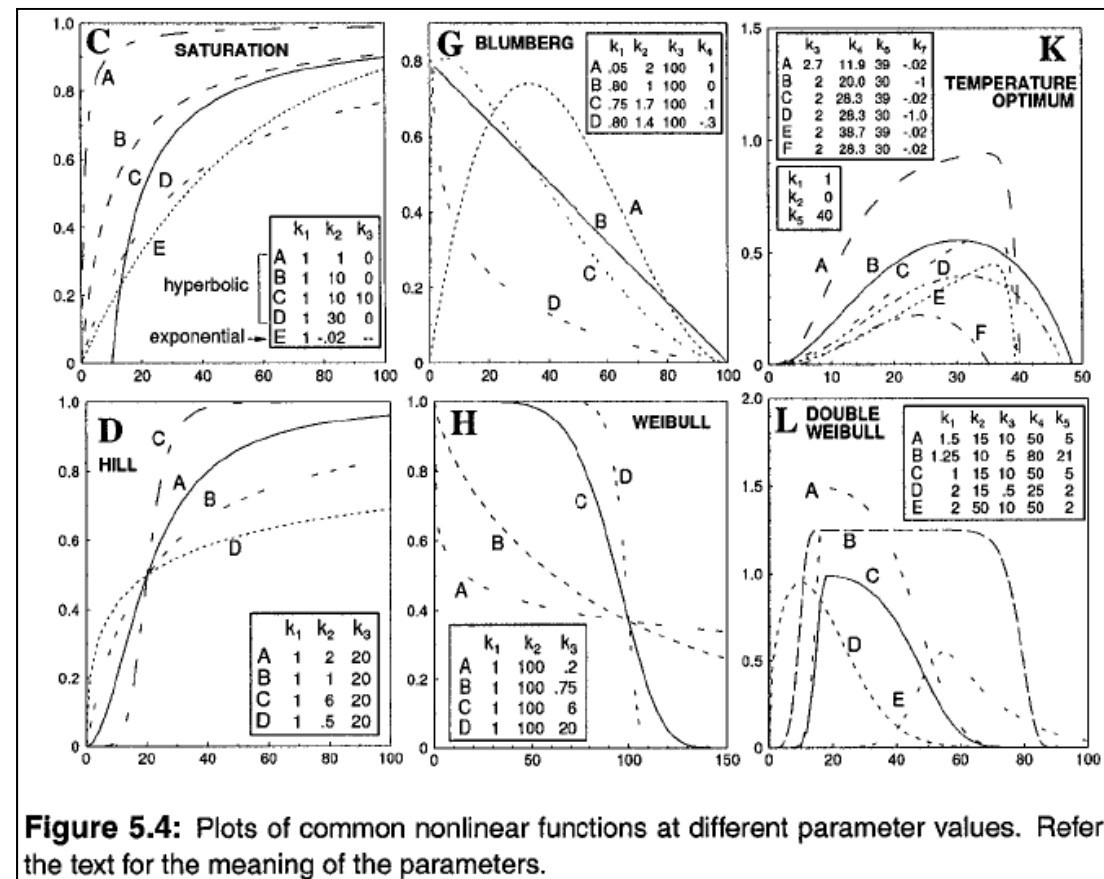
5.3 USEFUL FUNCTIONS

Summary of Useful Functions



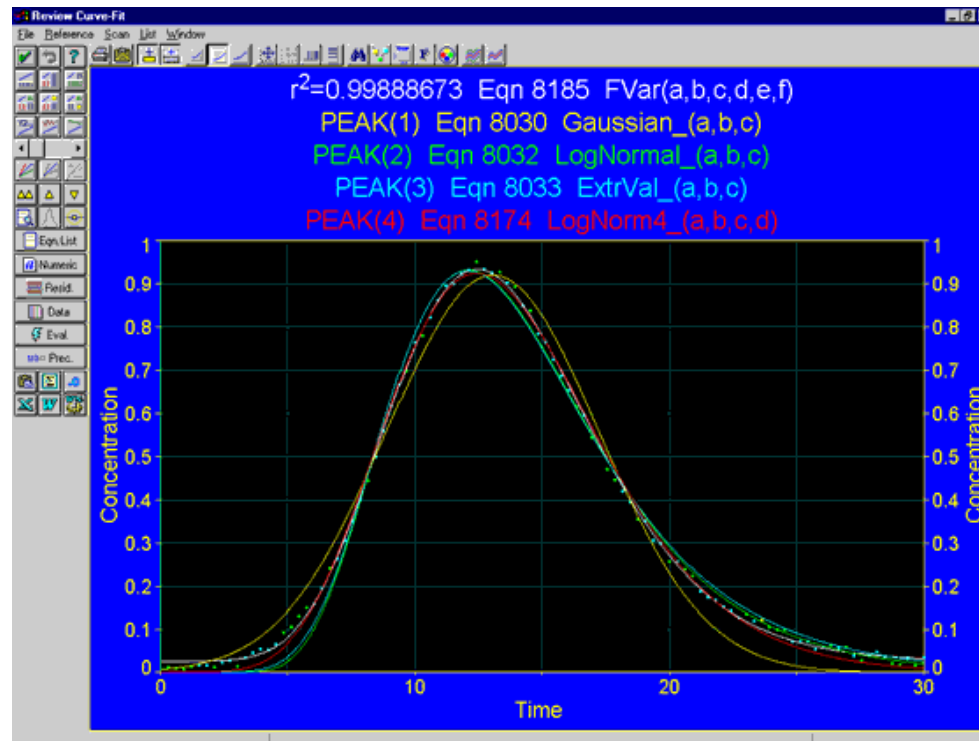
5.3 USEFUL FUNCTIONS

Summary of Useful Functions



5.3 USEFUL FUNCTIONS

- Using software to find suitable functions



<http://www.sigmaplot.co.uk/products/tablecurve2d/tablecurve2d.php>

<https://simfit.uk/simfit.html>



5.3 USEFUL FUNCTIONS

◎ Function Categories

◎ Linear Equations

- Polynomial Equations
- Rational Equations
- High Precision Polynomials and Rationals
- Chebyshev Series Equations
- Fourier Series Equations
- Constrained Non-Linear Rationals

◎ Non-linear Equations

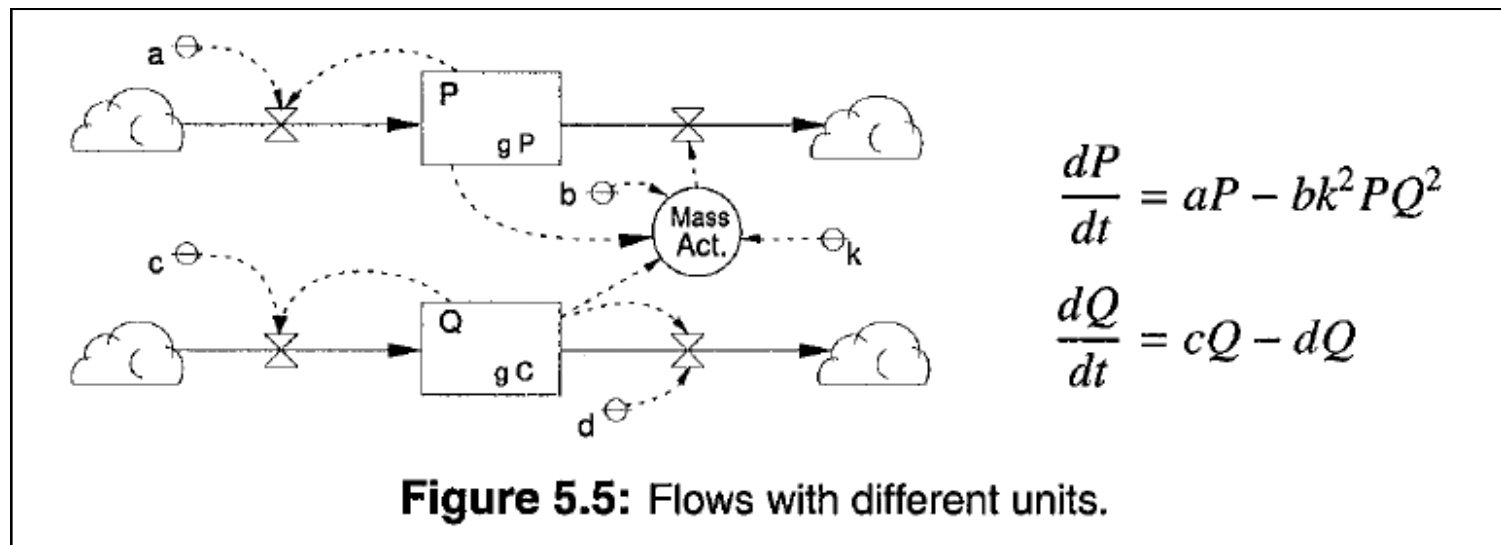
- Peak Functions
- Transition Functions
- Kinetic Functions
- Waveform Equations
- General Non-Linear Equations



5.4 EXAMPLES

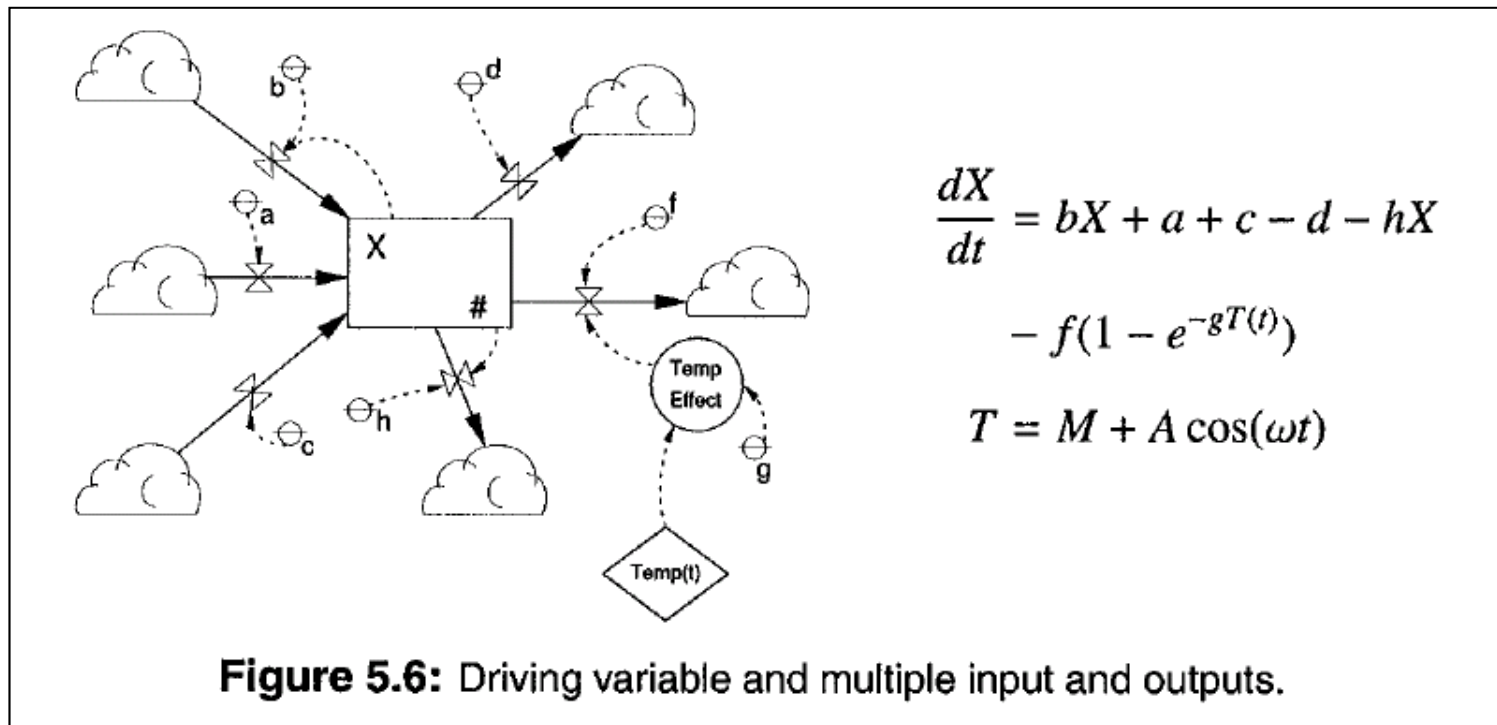
Flows with Different Units

Since the state variables have different units, we must use a parallel model, with information flows between state variables and rates to indicate the interactions.



5.4 EXAMPLES

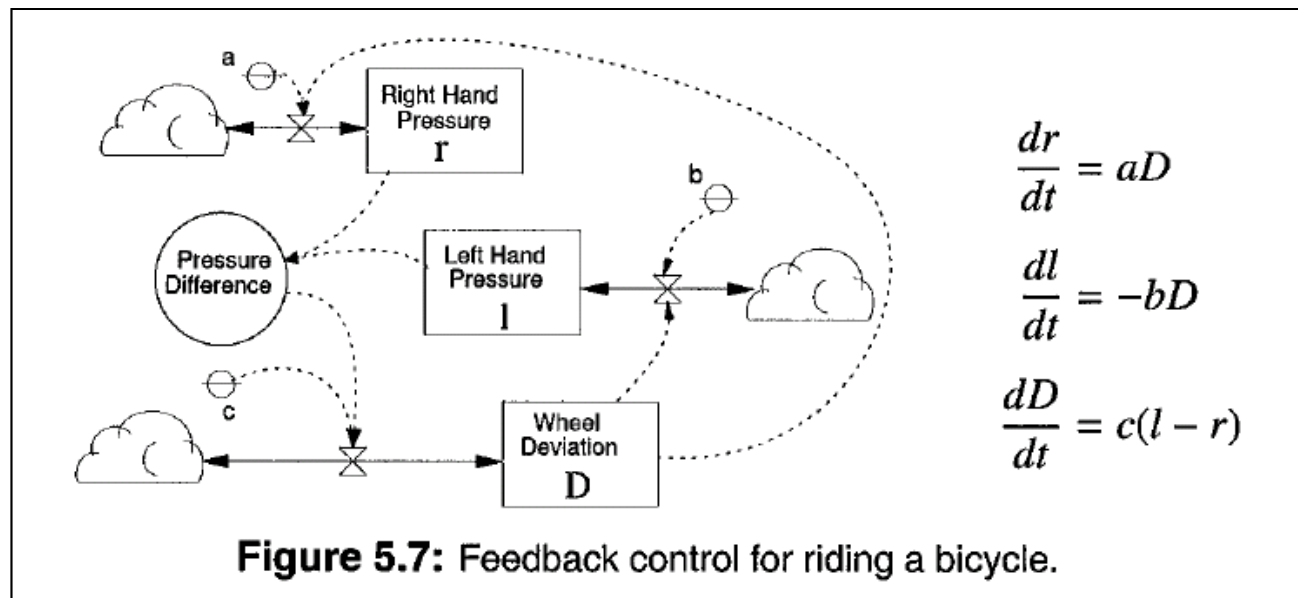
☉ Driving Variable



5.4 EXAMPLES

🎯 Riding a Bike

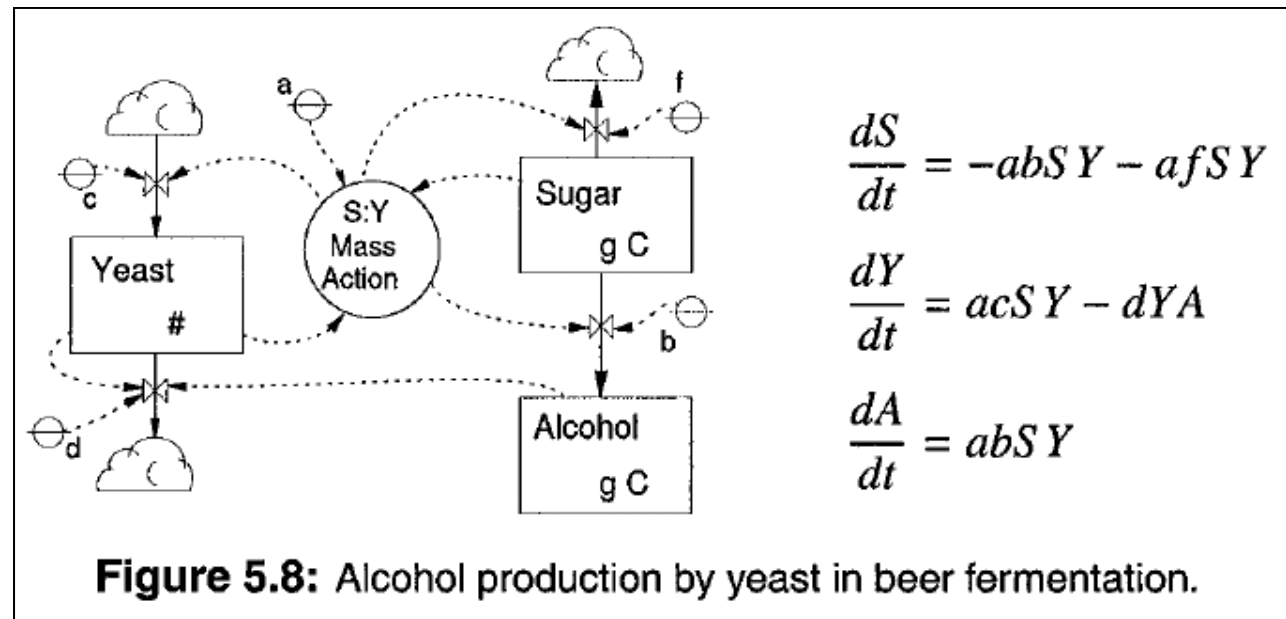
The problem is to describe the dynamics of the front wheel of a two-wheeled bicycle when it is driven (a) with hands in the normal position (left hand on the left handlebar, right hand on the right handlebar) and (b) with hands reversed.



5.4 EXAMPLES

☉ Brewing Beer

- ☉ There is only a finite amount of sugar at the beginning and it is depleted over time.
- ☉ Excessive alcohol will kill yeast cells.



5.4 EXAMPLES

☉ Brewing Beer

☉ Vensim Model

